

Introduction to Data Science and Engineering

- Probability



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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Some slides retrofitted by Michael Wu



You might want to go over Chapter 12 summary statistics yourself as a revision to what we have mentioned or covered in the data visualization lectures

<http://rafalab.dfci.harvard.edu/dsbook/summary-statistics.html>



In this lecture

- Probability and random numbers
- Discrete probability
- Independence and conditional probability
- Continuous probability: expected value and variance
- The Law of Large Numbers, Central Limit Theory
- Case study: bank loan
- The debate between Markov and Nekrasov

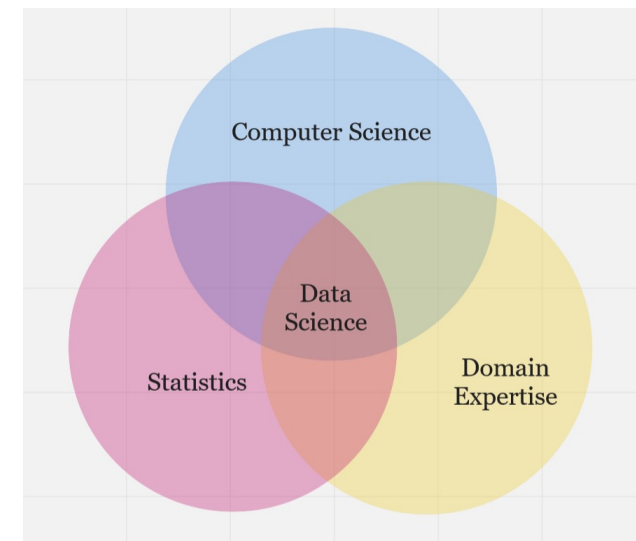


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What is Probability?

- Outcome of flipping a coin
- Chances being stuck in the MTR elevator queue
- Mentioning a word
- Mentioning a word “a lot”



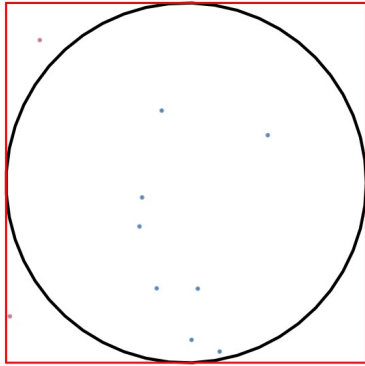


What is Probability?

- The chance that something will happen
- Given **infinite observations** of an event, the proportion of observations where a given outcome happens
- Strength of belief that something is true
- *“Probability is a mathematical language for quantifying uncertainty”* - Larry Alan Wasserman

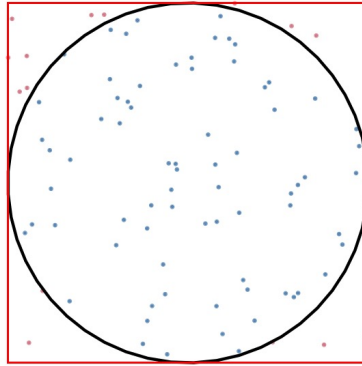
How to calculate π ?

10 POINTS



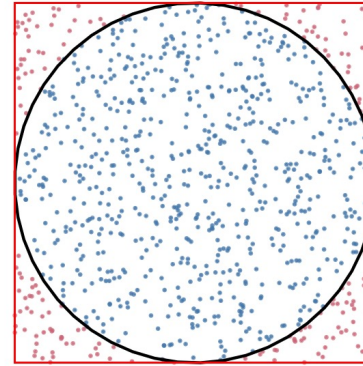
$\pi = 3.2$

100 POINTS



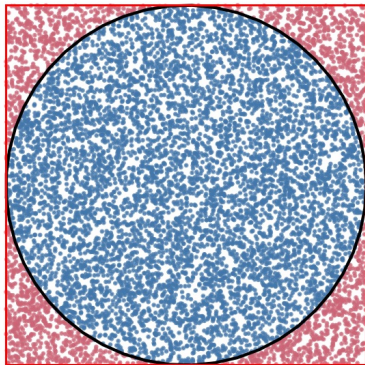
$\pi = 3.28$

1000 POINTS



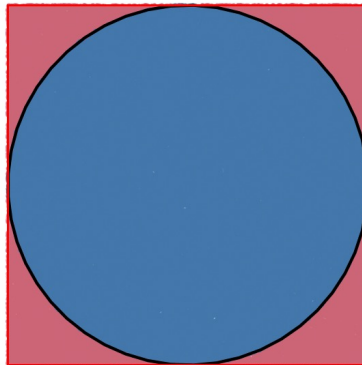
$\pi = 3.192$

10000 POINTS



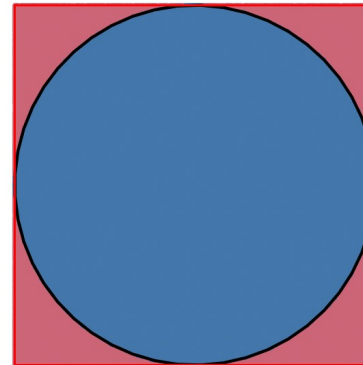
$\pi = 3.1216$

100000 POINTS



$\pi = 3.14452$

1000000 POINTS



$\pi = 3.14197$

There are tens of ways to calculate π , the method on the left that leverages probability theory, is called Monte Carlo simulation, or Monte Carlo method

Use Monte Carlo method to estimate π in R

```
estimate_pi <- function(n_iter, seed=42){  
  set.seed(seed)  
  
  x <- runif(n_iter, min=0, max=1) - 0.5  
  y <- runif(n_iter, min=0, max=1) - 0.5  
  dist <- sqrt(x**2 + y**2)  
  area <- mean(dist <= 0.5)  
  
  pi <- 4 * area  
  pi  
}
```

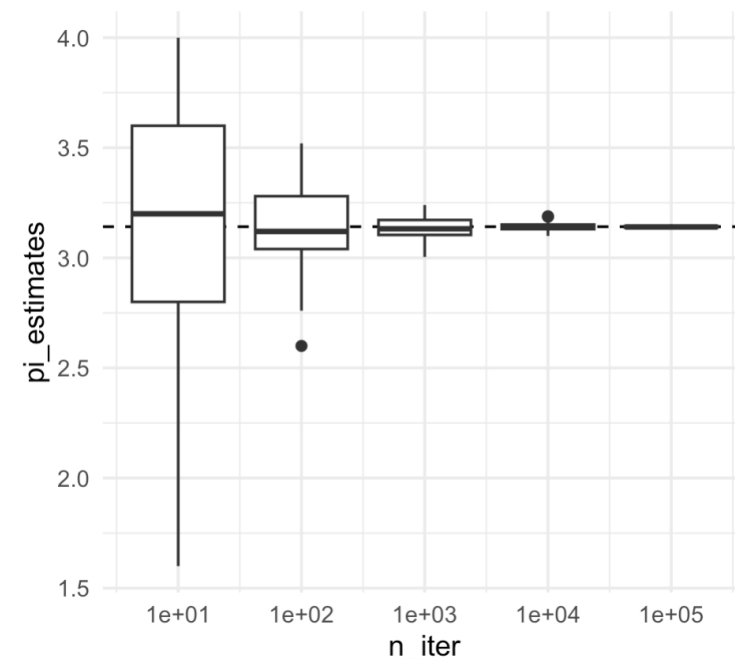
```
> n_iters <- c(1e1, 1e2, 1e3, 1e4, 1e5)  
> seeds <- seq(1:100)  
> df <- expand.grid(n_iters, seeds) |> setNames(c("n_iter", "seed"))  
> df$pi_estimates <- mapply(estimate_pi, df$n_iter, df$seed)
```

- `runif` is not run-if, it is r-unif
 - Sampling from a uniform distribution between `min` and `max`
- `set.seed` for setting the random seed

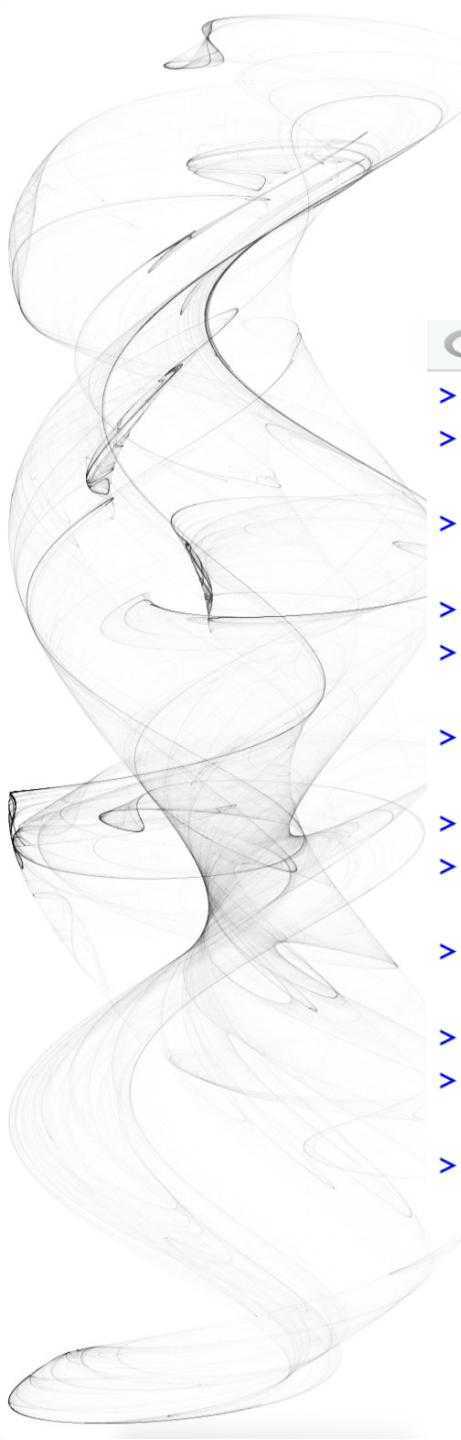
Use Monte Carlo method to estimate π in R

```
> df |>
+   group_by(n_iter) |>
+   summarize(mean=mean(pi_estimates), sd=sd(pi_estimates)) |>
+   mutate(mean=format(mean, digits=5))
# A tibble: 5 × 3
  n_iter mean      sd
  <dbl> <chr>    <dbl>
1     10 3.1560 0.533
2    100 3.1336 0.180
3   1000 3.1349 0.0518
4  10000 3.1415 0.0173
5 100000 3.1410 0.00481
```

```
> df |>
+   ggplot(aes(x = n_iter, y = pi_estimates, group = n_iter)) +
+   geom_hline(yintercept = pi, linetype = "dashed") +
+   geom_boxplot() +
+   scale_x_log10() +
+   theme_minimal()
```



Random Number Generator (RNG)



```
R 4.2.2 · ~/ 
> set.seed(42)
> round(runif(10,1,10))
[1] 9 9 4 8 7 6 8 2 7 7
> round(runif(10,1,10))
[1] 5 7 9 3 5 9 10 2 5 6
> set.seed(42)
> round(runif(10,1,10))
[1] 9 9 4 8 7 6 8 2 7 7
> round(runif(10,1,10))
[1] 5 7 9 3 5 9 10 2 5 6
> set.seed(21)
> round(runif(10,1,10))
[1] 8 3 7 3 10 9 2 3 10 9
> round(runif(10,1,10))
[1] 7 9 2 7 3 1 6 4 5 7
> set.seed(42)
> round(runif(10,1,10))
[1] 9 9 4 8 7 6 8 2 7 7
> round(runif(10,1,10))
[1] 5 7 9 3 5 9 10 2 5 6
```

- After setting a seed, the random number output sequence is certain
- Resetting with the same seed, the sequence starts again
- Setting a different seed gives a different sequence
- Use the same seed if the results requires to be repeatable
- Keep your seed a secret if the random number sequence is a secret
- RNG that requires a seed to start with is called pseudo RNG

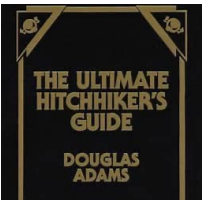


How to get true random number?
Pick a random number between 0-100

Pick a random number between 0-100



- Top picks
 - 73
 - 77
 - 37
 - 69
 - 42 (the answer to everything)
 - 7



The Hitchhiker's Guide to the Galaxy, Douglas Adams

Veritasium: <https://www.youtube.com/watch?v=d6iQrh2TK98&t=907s>



PseudoRandom Number Generator

- One example: linear congruential generator

The generator is defined by the **recurrence relation**:

$$X_{n+1} = (aX_n + c) \bmod m$$

where X is the **sequence** of pseudo-random values, and

m , $0 < m$ — the "**modulus**"

a , $0 < a < m$ — the "multiplier"

c , $0 \leq c < m$ — the "increment"

X_0 , $0 \leq X_0 < m$ — the "seed" or "start value"

(True) Random Number Generator

- Need to base on actual physical process:
 - thermal noises, signal jitter, Brownian motion, quantum phenomena, nuclear decay, etc.
- Rate-limited, cannot generate random numbers at high throughput. Why?

Article | [Open access](#) | Published: 09 August 2021

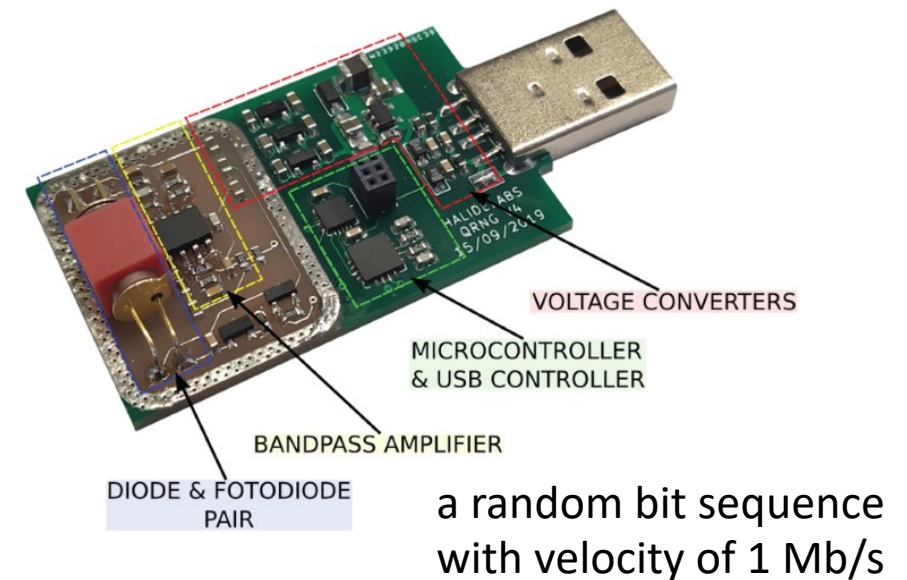
Quantum generators of random numbers

[Marcin M. Jacak](#), [Piotr Jóźwiak](#), [Jakub Niemczuk](#) & [Janusz E. Jacak](#) 

[Scientific Reports](#) **11**, Article number: 16108 (2021) | [Cite this article](#)

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Jacak, Marcin M., et al. "Quantum generators of random numbers." *Scientific Reports* 11.1 (2021): 16108.





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Discrete Probability

- Outcomes when using Monte Carlo simulation to calculate π :
 - The point is within the circle
 - Probability is $[\text{area of the circle}]/[\text{area of the square envelope}]$
 - The point is not in the circle
 - Probability is $1 - [\text{area of the circle}]/[\text{area of the square envelope}]$



Discrete Probability

- 5 beads in a pot:

```
> beads <- rep(c("red", "blue"), times = c(2,3))  
> beads  
[1] "red"  "red"  "blue" "blue" "blue"
```

- Pick a bead:

```
> sample(beads, 1)  
[1] "blue"
```

```
> mean(replicate(1e5, sample(beads, 1) == "blue"))  
[1] 0.60048
```




Notations

- Ω : Sample Space, set of all outcomes of a random experiment
- A : Event ($A \subseteq \Omega$), collection of possible outcomes of an experiment
- $P(A)$: Probability of event A , P is a function

Properties:

$$P(\Omega) = 1$$

$$P(A) \geq 0, \text{ for all } A$$

If $A_1, A_2, \dots$ are independent events then

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$$

$$\text{i.e., } P(A, B, \dots, X) = P(A) * P(B) * \dots * P(X)$$



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Independence

- We say two events are independent if the outcome of one does not affect the other
- Are the events independent in
 - Outcome of flipping a coin (two sequential trials)
 - Chances being stuck in the MTR elevator queue (Monday / Tuesday)
 - If there is a storm signal No.8 (next Monday / next Tuesday)
 - Keep unfolding cards from a deck, the chance of getting an Ace
 - Schrödinger's (two) cat(s)



Other probability properties

- If $B \subseteq A$ then $P(A) \geq P(B)$
- $P(\neg A) = 1 - P(A)$
- $P(A \cup B) \leq P(A) + P(B)$
 - Two events, A and B, are independent iff $P(A \cup B) = P(A) + P(B)$
- $P(A, B) \leq \min(P(A), P(B))$
 - Two events, A and B, are independent iff $P(A, B) = P(A) \times P(B)$



Gonna prove this
through conditional
probabilities



Independence

- For the five-bead example, are the first draw and the second draw independent?

```
> p <- mean(replicate(1e5, all(sample(beads, 1) == "blue")))
> p ** 2
[1] 0.36142
```

```
> mean(replicate(1e5, all(sample(beads, 2) == "blue")))
[1] 0.2986
```



Independence

- Are the events independent in
 - Outcome of flipping two coins;
 - Chances being stuck in the MTR elevator queue (Monday / Tuesday);
 - If there is a storm signal No.8 (next Monday / next Tuesday);
 - Keep unfolding cards from a deck, the chance of getting an Ace;
 - Schrödinger's cat(s) in two experiments on the same day.



Independence

- Are the events independent in
 - Outcome of flipping two coins;
 - Chances being stuck in the MTR elevator queue (Monday / Tuesday);
 - If there is a storm signal No.8 (next Monday / next Tuesday);
 - Keep unfolding cards from a deck, the chance of getting an Ace;
 - Schrödinger's cat(s) in two experiments on the same day.
- Some common mistakes:
 - I have lost 10 times in a random game (flipping coin), I will win next time.
 - Results from an online survey including 1M participants are unbiased.
 - The 2008 financial crisis: more on that later



Conditional probabilities

- When events are not independent, conditional probabilities are useful

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

$P(\text{Getting an Ace} \mid \text{Four Ace on the table already}) = 0$

Another example:

$P(A)$: mentioning "happy" in a message = .01

$P(B)$: mentioning "birthday" in a message = .001

$P(A, B)$: = .0005

$$P(A|B) = \frac{.0005}{.001} = 0.5$$



Reconciling independence and conditional probability

$P(A)$: first flip of a fair coin is heads = .5

$P(B)$: second flip of the same coin is heads = .5

$P(A, B)$: = .25

$$P(A|B) = \frac{0.25}{0.5} = 0.5 = P(A)$$

Two events, A and B , are independent iff $P(A, B) = P(A) \times P(B)$

$P(A, B) = P(A) \times P(B)$ iff $P(A|B) = P(A)$

i.e., Observing B has no effect on probability of A



What's underlying LLMs

Improving Language Understanding by Generative Pre-Training

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Given an unsupervised corpus of tokens $\mathcal{U} = \{u_1, \dots, u_n\}$, we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

Calculating combinations and permutations

- What is the probability that if I draw five cards without replacement, I get all cards of the same suit, what is known as a “flush” in poker?
- In math courses you learned theory on how to make these computations. Here we focus on how to use R code to compute the answers to an easier question: what’s the probability of a King in the second hand provided that the first hand is also a King

To generate a deck of cards:

```
R 4.2.2 · ~/ ↵  
> suits <- c("Diamonds", "Clubs", "Hearts", "Spades")  
> numbers <- c("Ace", "Deuce", "Three", "Four", "Five", "Six", "Seven",  
+             "Eight", "Nine", "Ten", "Jack", "Queen", "King")  
> deck <- expand.grid(number=numbers, suit=suits)  
> deck <- paste(deck$number, deck$suit)
```

```
> deck  
[1] "Ace Diamonds" "Deuce Diamonds" "Three Diamonds" "Four Diamonds"  
[5] "Five Diamonds" "Six Diamonds" "Seven Diamonds" "Eight Diamonds"  
[9] "Nine Diamonds" "Ten Diamonds" "Jack Diamonds" "Queen Diamonds"  
[13] "King Diamonds" "Ace Clubs" "Deuce Clubs" "Three Clubs"  
[17] "Four Clubs" "Five Clubs" "Six Clubs" "Seven Clubs"  
[21] "Eight Clubs" "Nine Clubs" "Ten Clubs" "Jack Clubs"  
[25] "Queen Clubs" "King Clubs" "Ace Hearts" "Deuce Hearts"  
[29] "Three Hearts" "Four Hearts" "Five Hearts" "Six Hearts"  
[33] "Seven Hearts" "Eight Hearts" "Nine Hearts" "Ten Hearts"  
[37] "Jack Hearts" "Queen Hearts" "King Hearts" "Ace Spades"  
[41] "Deuce Spades" "Three Spades" "Four Spades" "Five Spades"  
[45] "Six Spades" "Seven Spades" "Eight Spades" "Nine Spades"  
[49] "Ten Spades" "Jack Spades" "Queen Spades" "King Spades"
```

Calculating combinations and permutations

```
R 4.2.2 · ~/ ↗  
> kings <- paste("King", suits)  
> mean(deck %in% kings)  
[1] 0.07692308
```

Permutation:

- 3,1 is different than 1,3
- (1,1), (2,2), and (3,3) do not appear



```
R 4.2.2 · ~/ ↗  
> library(gtools)  
> permutations(3, 2)  
      [,1] [,2]  
[1,]    1    2  
[2,]    1    3  
[3,]    2    1  
[4,]    2    3  
[5,]    3    1  
[6,]    3    2
```

```
R 4.2.2 · ~/ ↗  
> hands <- permutations(52, 2, v = deck)
```

With the deck constructed, we can double check that the probability of a King in the first card is 1/13 by computing the proportion of possible outcomes that satisfy our condition

How about the conditional probability of the second card being a King given that the first was a King? If one King is already out of the deck and there are 51 left, then this probability is 3/51. Let's confirm by listing out all possible outcomes. we can use the permutations function from the gtools package

Permutation accepts a vector v. To compute all possible ways we can choose two cards when the order matters

Calculating combinations and permutations

```
R 4.2.2 · ~/ 
> kings <- paste("King", suits)
> sum(hands[,1] %in% kings)
[1] 204
```

```
R 4.2.2 · ~/ 
> sum(hands[,1] %in% kings & hands[,2] %in% kings) / sum(hands[,1] %in% kings)
[1] 0.05882353
```

$$P(B|A) = \frac{P(A, B)}{P(A)}$$

Now the cases for which the first hand was a King can be computed like this

To get the conditional probability, we compute what fraction of these have a King in the second card, which is exactly 3/51

Calculating combinations and permutations

```
R 4.2.2 · ~/ 
> combinations(3,2)
      [,1] [,2]
[1,]    1    2
[2,]    1    3
[3,]    2    3
```

How about if the order doesn't matter? For example, in Blackjack if you get an Ace and a face card in the first draw, it is called a Natural 21 and you win automatically. If we wanted to compute the probability of this happening, we would enumerate the combinations, not the permutations, since the order does not matter

```
R 4.2.2 · ~/ 
> aces <- paste("Ace", suits)
>
> facecard <- c("King", "Queen", "Jack", "Ten")
> facecard <- expand.grid(number = facecard, suit = suits)
> facecard <- paste(facecard$number, facecard$suit)
>
> hands <- combinations(52, 2, v = deck)
> mean((hands[,1] %in% aces & hands[,2] %in% facecard) |
+      (hands[,2] %in% aces & hands[,1] %in% facecard))
[1] 0.04826546
```

To compute the
probability of a Natural
21 in Blackjack



Monte Carlo method to estimate the probability of a Natural 21

```
R 4.2.2 · ~/ ➤  
> blackjack <- function(){  
+   hand <- sample(deck, 2)  
+   (hand[1] %in% aces & hand[2] %in% facecard) |  
+   (hand[2] %in% aces & hand[1] %in% facecard)  
+ }  
> rep <- 1e6  
> results <- replicate(rep, blackjack())  
> mean(results)  
[1] 0.048522
```

Discussion:

When to use Monte Carlo simulation (best-so-far solution)?

When to figure out the optimal solution?

How many times of experiments in Monte Carlo simulation is considered enough?



Other well-known examples

- The Monty Hall problem
- The birthday paradox
 - In a group of 50 people, there's a 96%+ chance that two people share the same birthday.



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Continuous probability

- Continuous probability focused on analyzing the distributions and observations of continuous variables
- What is the proportion of students in HKU that have heights of 170cm?
- What is the proportion of students in HKU that have heights of 170.0000000000cm?



Continuous probability properties

- We call X is a continuous random variable if it can take on an infinite number of values between any two given values
- X : observation $X(\omega)$ on inches of snowfall in a snowstorm event ω
- What is the probability we receive (at least) $\mathbf{a}$ inches?
 - $P(X \geq \mathbf{a}) := P(\{\omega: X(\omega) \geq \mathbf{a}\})$
- What is the probability we receive between $\mathbf{a}$ and $\mathbf{b}$ inches?
 - $P(\mathbf{a} \leq X \leq \mathbf{b}) := P(\{\omega: \mathbf{a} \leq X(\omega) \leq \mathbf{b}\})$
- Probability of receiving exactly i inches of snowfall is zero
 - $P(X = i) := 0$, for all $i \in \Omega$

Cumulative distribution functions (CDF)

```
R 4.2.2 · ~/ ↵  
> library(tidyverse)  
> library(dslabs)  
> data(heights)  
> x <- heights |> filter(sex=="Male") |> pull(height)  
> F <- function(a) mean(x<=a)
```

```
R 4.2.2 · ~/ ↵  
> 1 - F(70.5)  
[1] 0.3633005
```

$$F(a) = \Pr(X \leq a)$$

$$\Pr(a < X \leq b) = F(b) - F(a)$$

We used the heights of adult male students as an example. We define the empirical cumulative distribution function (eCDF) as F which, for any value a , gives the proportion of values in the list x that are smaller or equal than a

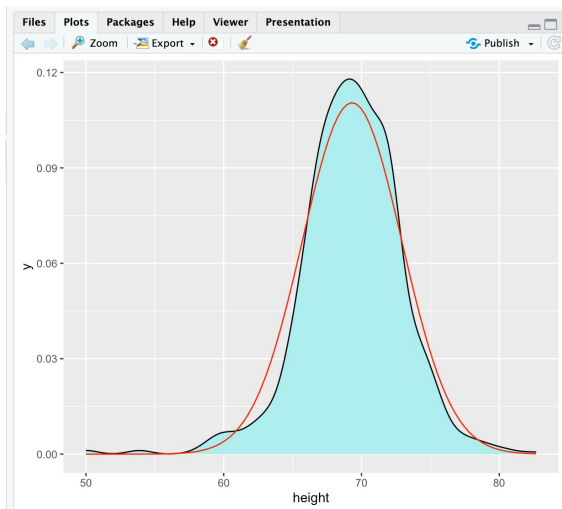
Let's connect the eCDF to probability by asking: if I pick one of the male students at random, what is the chance that he is taller than 70.5 inches? Because every student has the same chance of being picked, the answer to this is equivalent to the proportion of students that are taller than 70.5 inches. Using the eCDF we obtain an answer by this

The CDF is a version of the eCDF that assigns theoretical probabilities for each a rather than proportions computed from data. Specifically, the CDF for a random outcome X defines, for any number a , the probability of observing a value that is larger than a

Once a CDF is defined, we can use it to compute the probability of any subset of values. Because we can compute the probability for any possible event this way, the CDF defines the probability distribution

Probability density function (PDF)

$$F(b) - F(a) = \int_a^b f(x) dx$$



```
R 4.2.2 · ~/
> ?pnorm
```

```
R 4.2.2 · ~/ ↵
> m <- mean(x)
> s <- sd(x)
> 1 - pnorm(70.5, m, s)
[1] 0.371369
```

A mathematical result that is actually very useful in practice is that for most CDFs we can define a function, call it $f(x)$, that permits us to construct the CDF using Calculus. $f(x)$ is referred to as the probability density function.

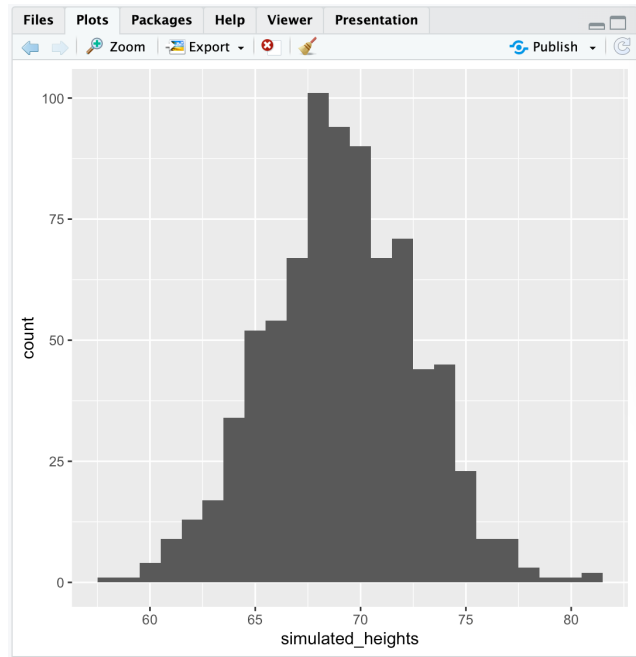
An example of such a continuous distribution is the normal distribution, the probability density function of a normal distribution is as below, where m is the mean, s is the sd:

$$f(x) = e^{-\frac{1}{2} \left(\frac{x-m}{s} \right)^2}$$

The cumulative distribution for the normal distribution is defined by a mathematical formula which in R can be obtained with the function `pnorm`

This is useful because if we are willing to use the normal approximation for, say, height, we don't need the entire dataset to answer questions such as: what is the probability that a randomly selected student is taller than 70 inches? We just need the average height and standard deviation

Monte Carlo simulations for continuous variables



R provides functions to generate normally distributed outcomes. Specifically, the `rnorm` function takes three arguments: size, average, and standard deviation and produces random numbers. Here is an example of how we could generate data that looks like our reported heights

The Normal Distribution

R 4.2.2 · ~/

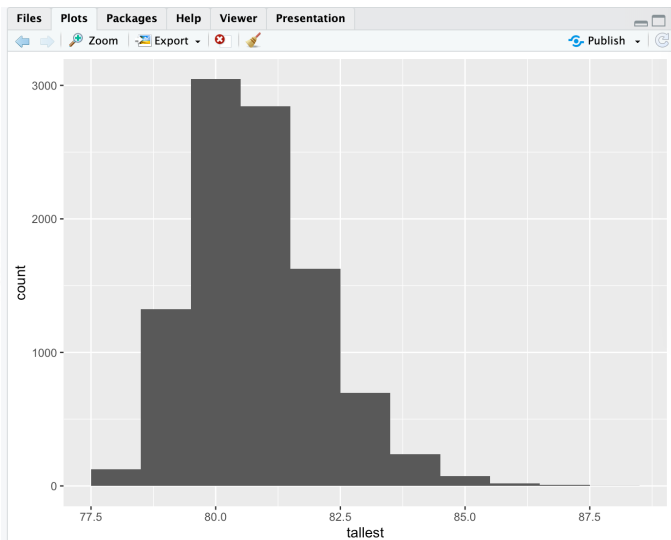
```
> n <- length(x)
> m <- mean(x)
> s <- sd(x)
> simulated_heights <- rnorm(n, m, s)
> as.data.frame(simulated_heights) |>
+ ggplot(aes(simulated_heights)) + geom_histogram(binwidth = 1)
```

Description

Density, distribution function, quantile function and random generation for the normal distribution with mean equal to mean and standard deviation equal to sd.

Usage

```
dnorm(x, mean = 0, sd = 1, log = FALSE)
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
rnorm(n, mean = 0, sd = 1)
```



If, for example, we pick 800 males at random, what is the distribution of the tallest person? The following Monte Carlo simulation helps us answer that question

R 4.2.2 · ~/

```
> rep <- 10000
> tallest <- replicate(rep, {
+   simulated_data <- rnorm(800, m, s)
+   max(simulated_data)
+ })
> as.data.frame(tallest) |>
+ ggplot(aes(tallest)) + geom_histogram(binwidth = 1)
```



- This is not a normal distribution
- These values are modeled through Gumbel distribution

https://en.wikipedia.org/wiki/Gumbel_distribution

Continuous distributions other than normal distribution

The normal distribution is not the only useful theoretical distribution. Other continuous distributions that we may encounter are the student-t, Chi-square, exponential, gamma, beta, and beta-binomial. R provides functions to compute the density, the quantiles, the cumulative distribution functions and to generate Monte Carlo simulations. R uses a convention that lets us remember the names, namely using the letters d, q, p, and r in front of a shorthand for the distribution.

- **d** → Density / Probability Mass Function (PDF/PMF)
- **p** → Cumulative Distribution Function (CDF)
- **q** → Quantile Function
- **r** → Random Number Generation

Example – Normal Distribution:

```
# PDF: Probability density at x = 0
dnorm(0, mean = 0, sd = 1)

# CDF: Probability that X ≤ 1.96
pnorm(1.96, mean = 0, sd = 1)

# Quantile: Value at the 97.5% percentile
qnorm(0.975, mean = 0, sd = 1)

# Random generation: 5 samples from N(0,1)
rnorm(5, mean = 0, sd = 1)
```

The Normal Distribution

Description

Density, distribution function, quantile function and random generation for the normal distribution with mean equal to mean and standard deviation equal to sd.

Usage

```
dnorm(x, mean = 0, sd = 1, log = FALSE)
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
rnorm(n, mean = 0, sd = 1)
```



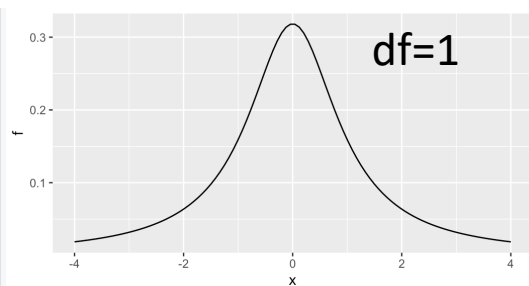
The Student t Distribution

Description

Density, distribution function, quantile function and random generation for the t distribution with df degrees of freedom (and optional non-centrality parameter ncp).

Usage

```
dt(x, df, ncp, log = FALSE)
pt(q, df, ncp, lower.tail = TRUE, log.p = FALSE)
qt(p, df, ncp, lower.tail = TRUE, log.p = FALSE)
rt(n, df, ncp)
```



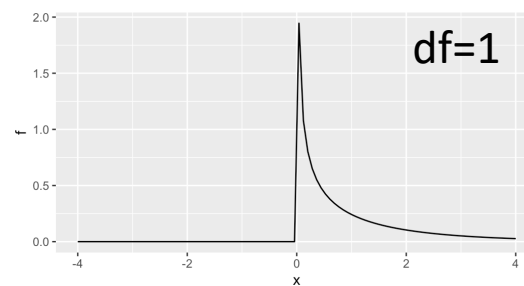
The (non-central) Chi-Squared Distribution

Description

Density, distribution function, quantile function and random generation for the chi-squared (χ^2) distribution with df degrees of freedom and optional non-centrality parameter ncp.

Usage

```
dchisq(x, df, ncp = 0, log = FALSE)
pchisq(q, df, ncp = 0, lower.tail = TRUE, log.p = FALSE)
qchisq(p, df, ncp = 0, lower.tail = TRUE, log.p = FALSE)
rchisq(n, df, ncp = 0)
```



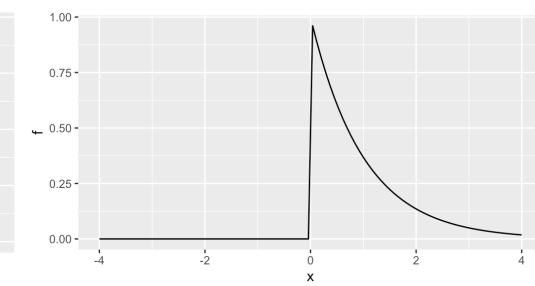
The Exponential Distribution

Description

Density, distribution function, quantile function and random generation for the exponential distribution with rate rate (i.e., mean 1/rate).

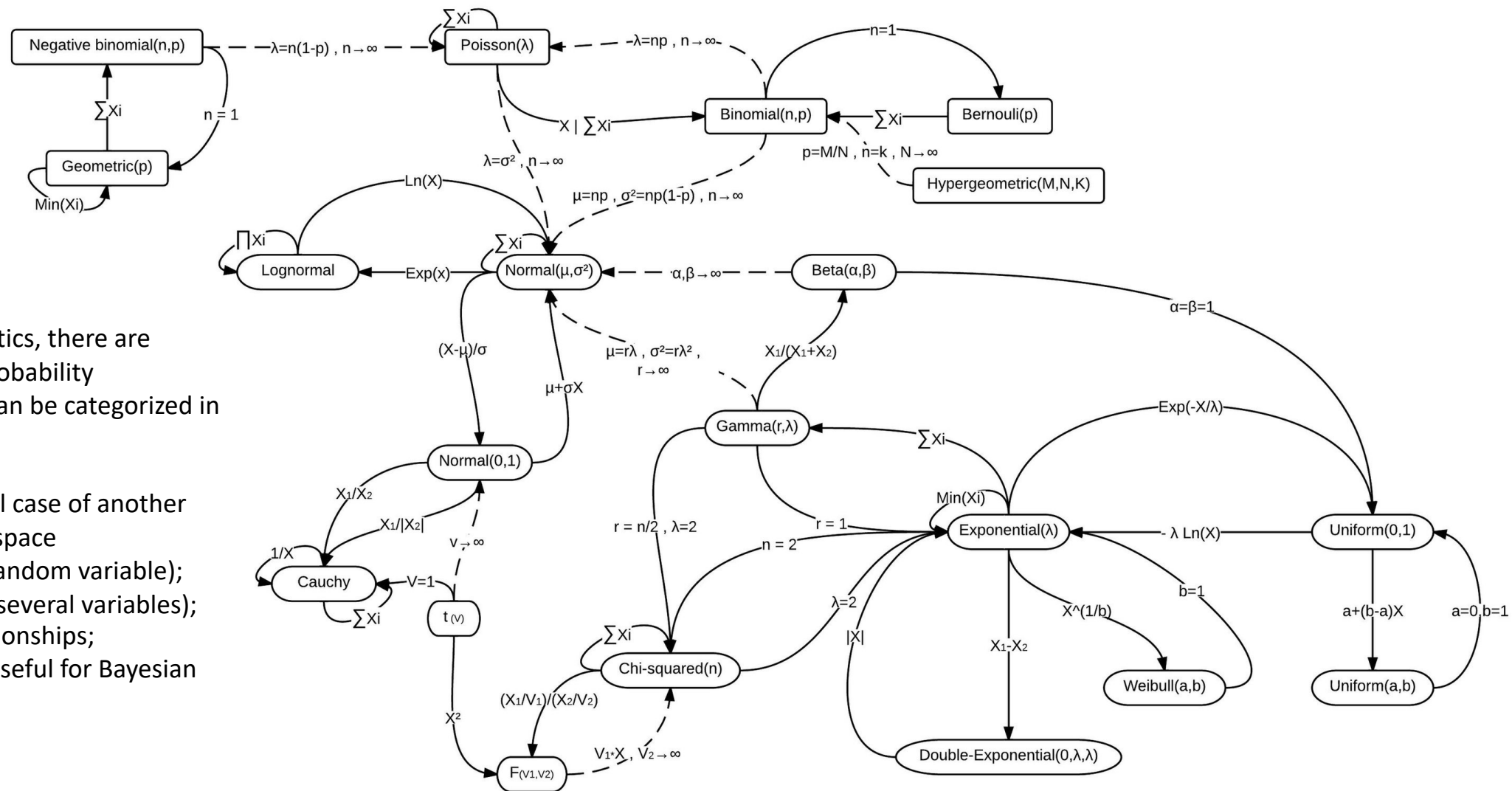
Usage

```
dexp(x, rate = 1, log = FALSE)
pexp(q, rate = 1, lower.tail = TRUE, log.p = FALSE)
qexp(p, rate = 1, lower.tail = TRUE, log.p = FALSE)
rexp(n, rate = 1)
```



...

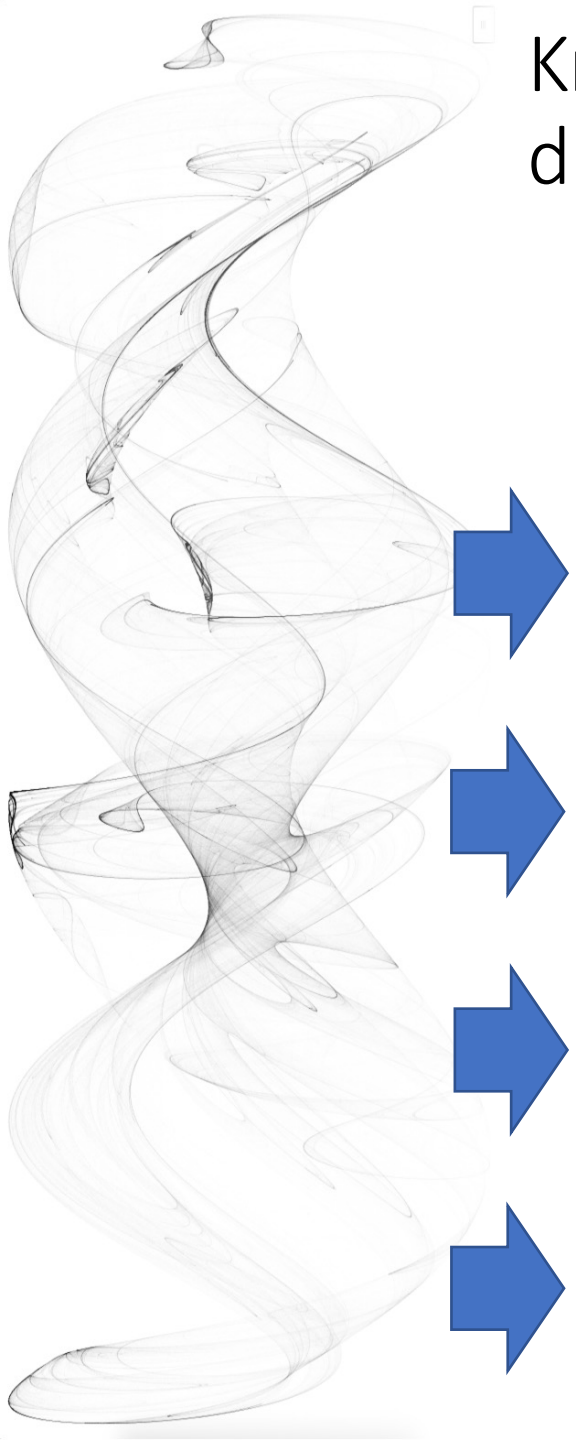
You gonna learn the relationship between the probability distributions in STAT2601/2



In probability theory and statistics, there are several relationships among probability distributions. These relations can be categorized in the following groups:

- One distribution is a special case of another with a broader parameter space
- Transforms (function of a random variable);
- Combinations (function of several variables);
- Approximation (limit) relationships;
- Compound relationships (useful for Bayesian inference);
- Duality;
- Conjugate priors.

Knowing what to learn about a probability distribution is good for now



	Normal Distribution	Student's T-Distribution	Binomial Distribution	Poisson Distribution	Exponential Distribution
What does it look like?					
Defining Characteristics	Distinctive Bell Shape	Shorter, fatter than the normal distribution.	Two outcomes: Success/Failure	Various shapes, but valid only for integers on the x-axis.	Models Time Between Events
Example of When to Use It	Modeling natural phenomena (height, weight, IQ, test scores etc.)	When you have small samples or don't know the population variance (σ^2).	Coin Toss Probability (Heads, Tails)	Gives probability of number of events in a fixed interval.	"How much time will go by before a major hurricane hits the Atlantic Seaboard?"
Example of DS Application	Least squares fitting or propagation of uncertainty.	Unknown σ^2 is common in real life data, you you'll have to use the T instead of the normal in that case.	Anywhere where binary (yes/no, black/white, vote/don't vote) data is used.	Anywhere there is a waiting time between events.	Building continuous-time Markov chains.



Expected value and variance

Two important properties for a random variable:

- Expected value: $E(X)$
 - Coin toss: head = 1, tail = -1; expected value is 0 in a perfect 50/50 scenario
 - Investment: weighted average of each possible outcome by its probability; expected value is the mean gain over many repetitions
- Variance: $\text{Var}(X) = E[(X - E(X))^2]$ and standard deviation $\text{SD}(X) = \sqrt{\text{Var}(X)}$
 - Investment: chance of getting high return or loss; measurement of risk
 - Tolerance (engineering): manufacture of the same item may have tiny variability in the end product; low tolerance = high quality



How to calculate them

Say random variable X may take values $x_1, x_2, \dots$

$$E(X) = \sum x_i P(x_i)$$

In real-world data, observations directly came from $P(X)$, the average of observations $\bar{X}$ converges to $E(X)$ as number of observations increase (the Law of Large Numbers).



In this lecture

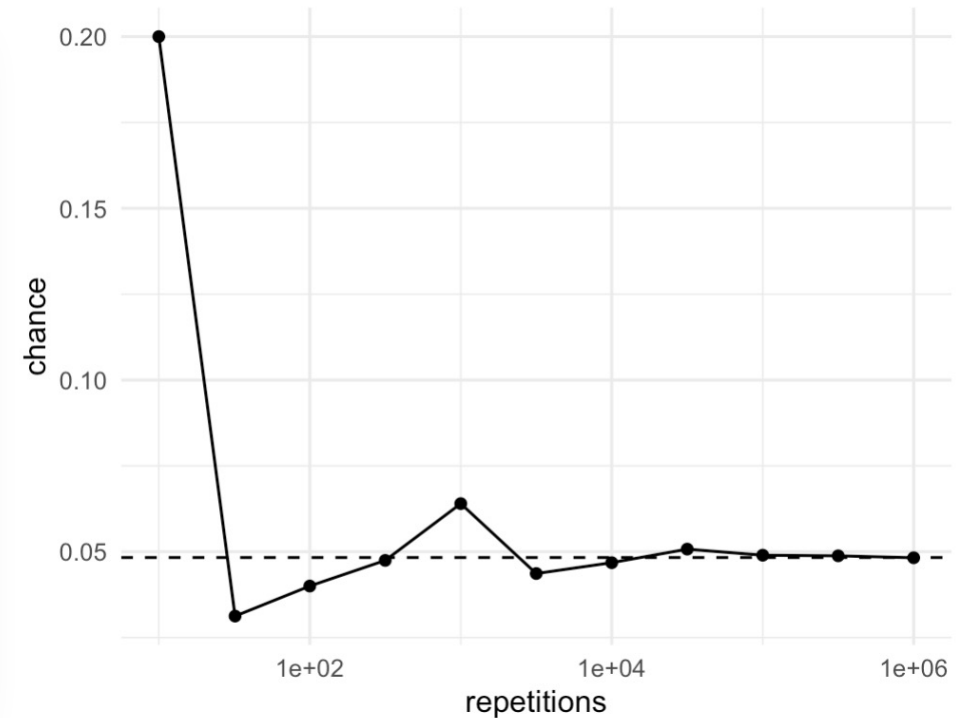
- Probability and random numbers
- Discrete probability
- Independence and conditional probability
- Continuous probability: expected value and variance
- **The Law of Large Numbers, Central Limit Theory**
- Case study: bank loan
- The debate between Markov and Nekrasov

The Law of Large Numbers

X : getting a natural 21 = 1, else = 0

$$E(X) = 0.0483$$

```
df <- data.frame(repetitions = round(10**seq(1, 6, 0.5)))
run_21_n_times <- function(n=10){
  outcome <- replicate(n, is_21())
  mean(outcome)
}
df <- df |> mutate(chance = sapply(repetitions, run_21_n_times))
df |>
  ggplot(aes(repetitions, chance)) +
    geom_line() +
    geom_point() +
    scale_x_log10() +
    geom_hline(yintercept=0.0483, lty="dashed") +
    theme_minimal()
```

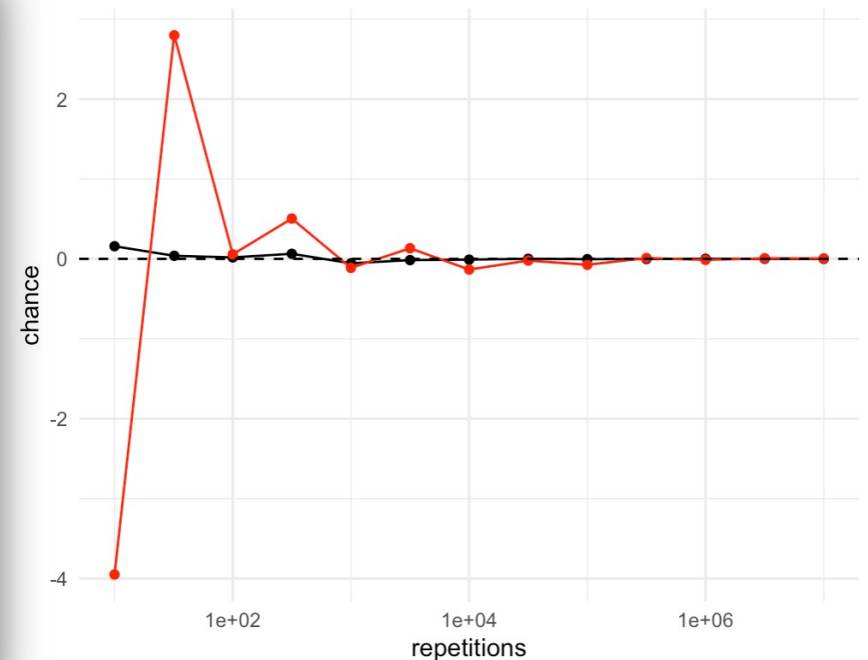


The Law of Large Numbers

X : sample a value from Gaussian distribution (mean = 0)

$$E(X) = 0$$

```
df <- data.frame(repetitions = round(10**seq(1, 7, 0.5)))
sample_standard_normal <- function(n=10, sd=1){
  outcome <- rnorm(n, mean = 0, sd = sd)
  mean(outcome)
}
df <- df |> mutate(chance = sapply(repetitions, sample_standard_normal))
df <- df |> mutate(chance_sd10 = sapply(repetitions, sample_standard_normal, sd=10))
df |>
  ggplot() +
    geom_line(aes(repetitions, chance)) +
    geom_point(aes(repetitions, chance)) +
    geom_line(aes(repetitions, chance_sd10, color="red")) +
    geom_point(aes(repetitions, chance_sd10, color="red")) +
    scale_x_log10() +
    geom_hline(yintercept=0., lty="dashed") +
    theme_minimal()
```





How to calculate them

Say random variable X may take values $x_1, x_2, \dots$

$E(X) = \sum x_i P(x_i)$ can be approximated by $\bar{X}$

Similarly, $SD(X) = \sqrt{\sum (x_i - E(X))^2 P(x_i)}$ can be approximated by:

$$\frac{1}{n} \sum (X_{obs} - \bar{X})^2$$

Because observations directly came from $P(X)$

- Divide by n or $n - 1$?
- For this class, we will use n if not otherwise specified

https://en.wikipedia.org/wiki/Bessel%27s_correction

Some important properties

- For n i.i.d. samples of X :

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$$

**distributive property
applies when i.i.d.
applies**

$$E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{E(X_1) + E(X_2) + \cdots + E(X_n)}{n} = E(X_1)$$

$$\text{Var}(X_1 + X_2 + \cdots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)$$

$$\text{Var}(CX_1) = \text{Var}(X_1) * C^2$$

$$\text{Var}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{\text{Var}(X_1) * n}{n^2} = \frac{\text{Var}(X_1)}{n}$$

$$\text{SD}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{\text{SD}(X_1)}{\sqrt{n}}$$

(this is often referred to as the standard error of sample mean)



Central Limit Theorem (CLT)

For random variable X that follows **ANY** distribution (with finite mean and variance), **sample mean** of n i.i.d. observations (this is also a random variable) will converge to:
a Gaussian distribution with $E(X)$ as mean, and $\frac{SD(X)}{\sqrt{n}}$ as sd, when **n is large**.



In this lecture

- Probability and random numbers
- Discrete probability
- Independence and conditional probability
- Continuous probability: expected value and variance
- The Law of Large Numbers, Central Limit Theory
- **Case study: bank loan**
- The debate between Markov and Nekrasov

Case study: bank loan

```
# Case study: loan
principal_per_loan <- 180000
operational_loss <- 20000
loss_per_foreclosure <- -(principal_per_loan + operational_loss)

sim_average_return <- function(interest_rate=0.02, default_rate=0.02, loans=1000){
  defaults <- sample(
    c(0,1), loans, prob=c(1-default_rate, default_rate), replace = TRUE)
  balance <- sum(defaults * loss_per_foreclosure) +
    sum((1 - defaults) * principal_per_loan * interest_rate)
  balance / loans
}
```

```
> set.seed(42)
> sim_average_return(0.02, 0.02)
[1] 749.6
> set.seed(43)
> sim_average_return(0.02, 0.02)
[1] -1286.4
```

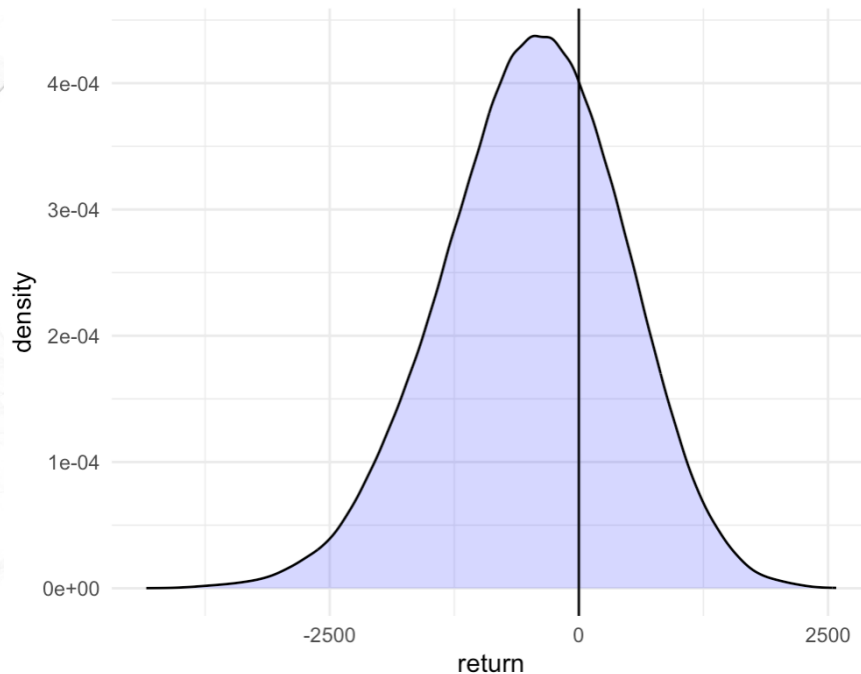
1,000 loans of \$180k. After adding up all costs, suppose your bank loses \$200,000 per foreclosure.

Assume that default rate is 2%:

- If defaulted, the bank will lose \$200,000
- If not, the bank will earn \$180k x interest rate

Outcome of each loan is a random variable;
`sim_average_return` is calculating its sample mean, which is also a random variable

```
return_sampled <- data.frame(  
  return=replicate(1e5, sim_average_return(0.02, 0.02))  
)  
return_sampled |>  
  ggplot(aes(return)) +  
  geom_density(adjust = 1.5, fill = "blue", alpha = 0.2) +  
  geom_vline(xintercept = 0) +  
  theme_minimal()
```



We can use Monte Carlo simulation to get an idea on the distribution of the average return

Takeaway:

- Sample mean (average return) is very “normal” (CLT);
- Majority of the returns are negative

```

return_sampled <- data.frame(
  interest_rate=rep(seq(0.02, 0.04, 0.001), 1e5))

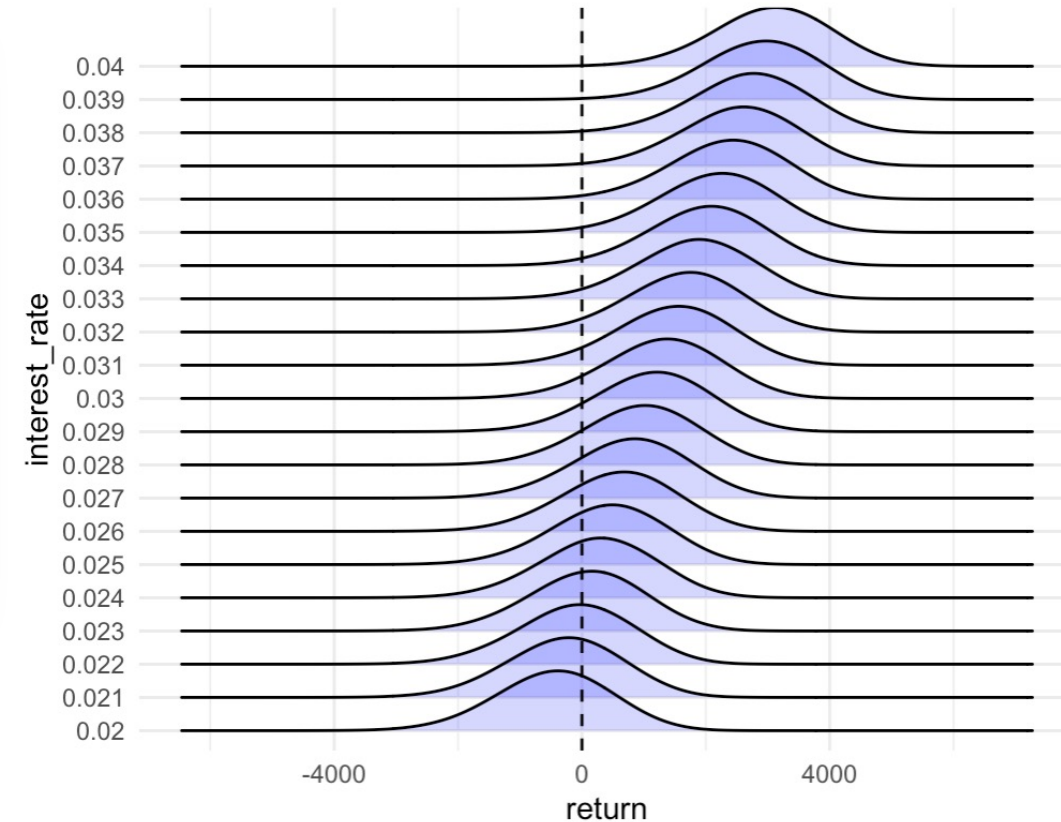
return_sampled <- return_sampled |>
  mutate(return=sapply(interest_rate, sim_average_return))

return_sampled |>
  mutate(interest_rate=as.factor(interest_rate)) |>
  ggplot(aes(return, interest_rate)) +
    geom_density_ridges(bandwidth=300, fill="blue", alpha=0.2) +
    geom_vline(xintercept=0., lty="dashed") +
    theme_minimal()

```

Banks will, therefore, increase the interest rate.

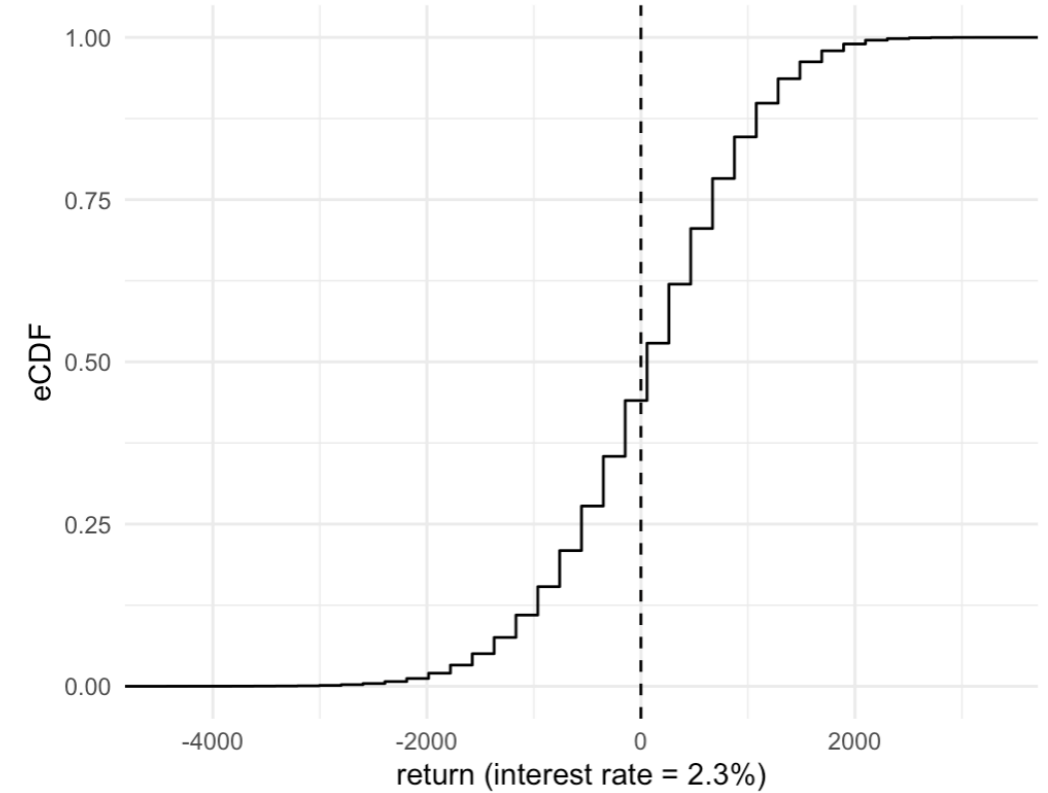
When interest rate surpasses 2.3%, the center of the distribution shifts to positive.




```
return_sampled |>
  filter(interest_rate==0.023) |>
  ggplot(aes(return)) +
    stat_ecdf(geom = "step", color = "black") +
    geom_vline(xintercept=0., lty="dashed") +
    theme_minimal() +
    xlab("return (interest rate = 2.3%)") +
    ylab("eCDF")
```

This is still not enough! We need an interest rate that lowers the chance of losing money to lower than 1%.

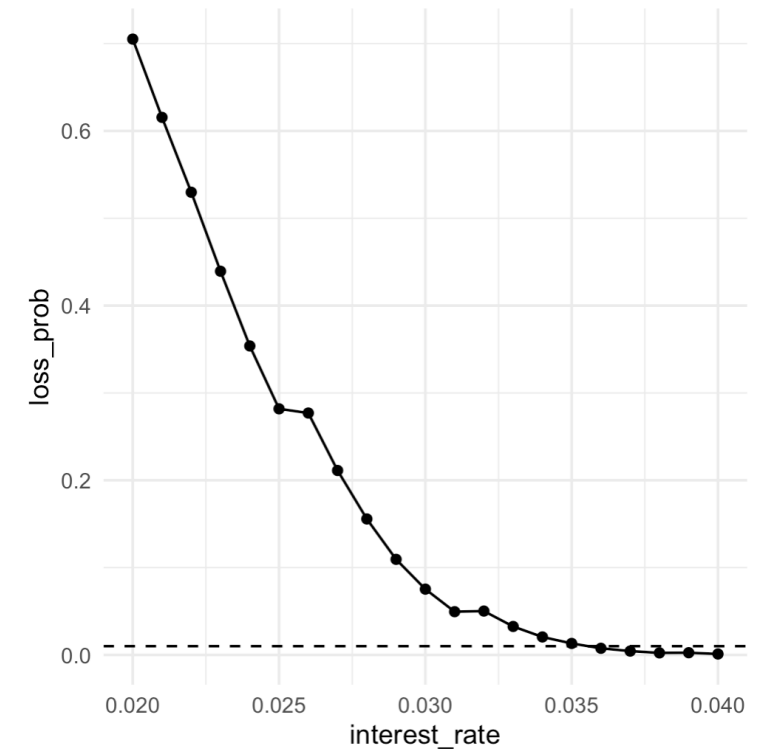
If interest rate = 2.3%, the bank has a ~50% chance of losing money




```
return_sampled |>
  group_by(interest_rate) |>
  summarize(loss_prob = mean(return < 0)) |>
  ggplot(aes(interest_rate, loss_prob)) +
    geom_point() +
    geom_line() +
    geom_hline(yintercept=0.01, lty="dashed") +
    theme_minimal()
```

One option is to keep increasing the interest rate.

At interest rate=3.5%, the chance of losing money is ~1%



```

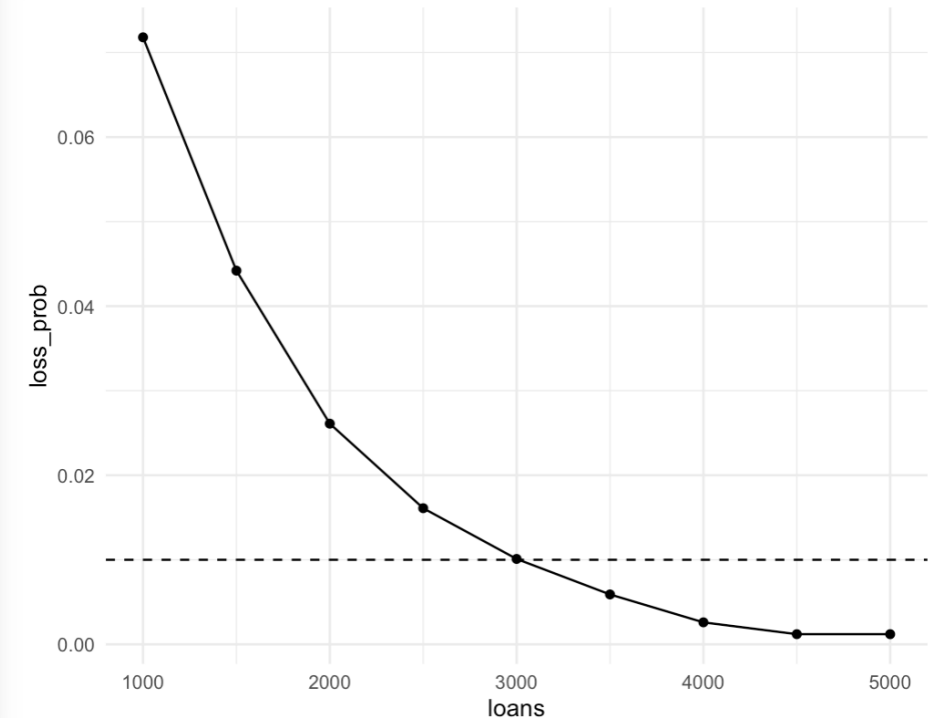
return_sampled <- data.frame(
  loans=rep(seq(1000, 5000, 500), 1e4))

f <- function(loans){
  sim_average_return(interest_rate=0.03, default_rate=0.02, loans=loans)
}

return_sampled <- return_sampled |>
  mutate(return=apply(loans, f))

return_sampled |>
  group_by(loans) |>
  summarize(loss_prob = mean(return < 0)) |>
  ggplot(aes(loans, loss_prob)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0.01, lty = "dashed") +
  theme_minimal()

```



Another option is to increase the size of loans (why?)

At interest rate=3%, giving out 3000 loans (instead of 1000) reduces losing chance to <1%

- Return is a sample mean, whose standard error will decrease if n increases.

```

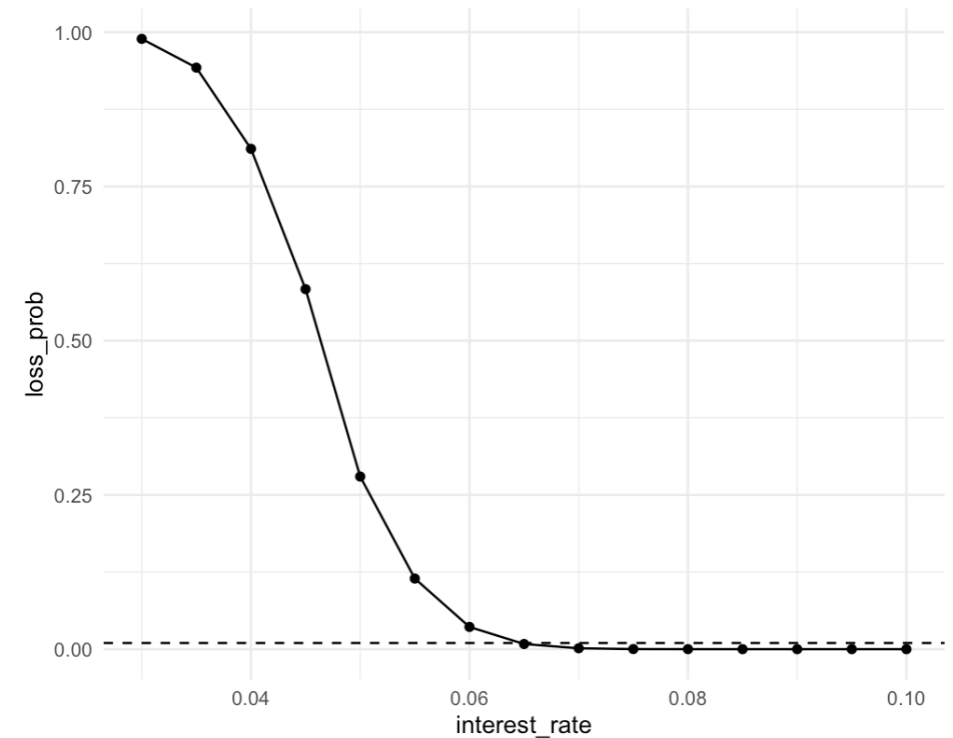
return_sampled <- data.frame(
  interest_rate=rep(seq(0.03, 0.1, 0.005), 1e5))

f <- function(i){
  sim_average_return(interest_rate=i, default_rate=0.04, loans=1000)
}

return_sampled <- return_sampled |>
  mutate(return=apply(interest_rate, f))

return_sampled |>
  group_by(interest_rate) |>
  summarize(loss_prob = mean(return < 0)) |>
  ggplot(aes(interest_rate, loss_prob)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0.01, lty = "dashed") +
  theme_minimal()

```



In economy recession, default rate might be higher than 2%;

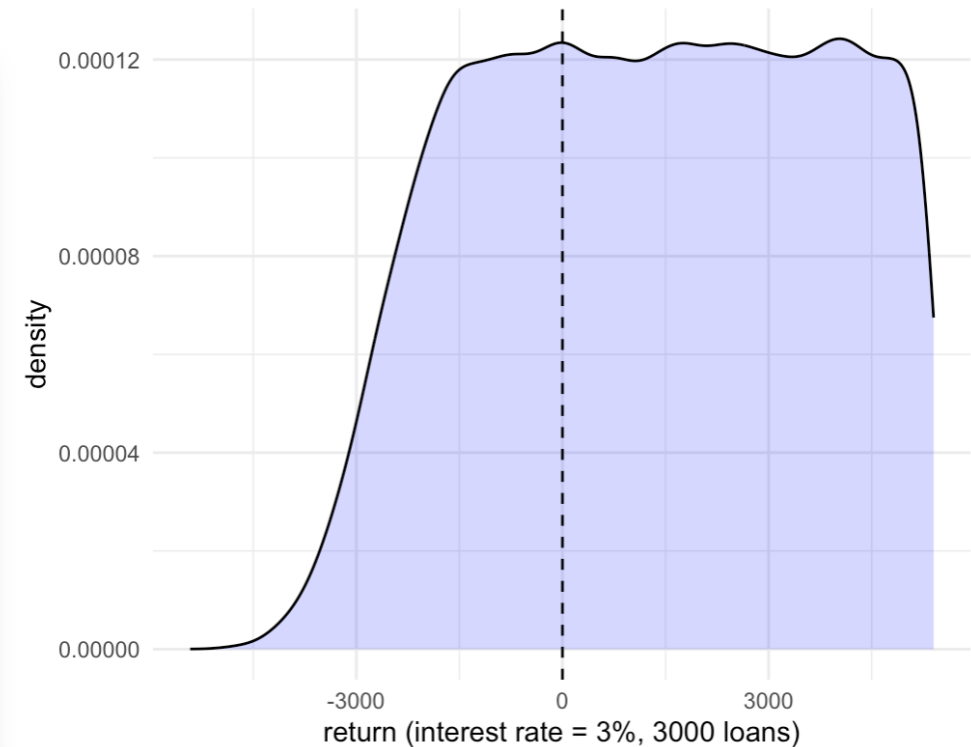
If default rate is at 4%, when giving out 1000 loans, the bank needs to have a 6.5+% interest rate to avoid losing

So far, we have assumed that the events of defaults are **independent** across loans, but is that true?

In economic crisis, everyone is affected, everyone will have a higher chance of bankruptcy.

```
sample_return_when_default_rate_fluc <- function(){  
  default_rate_fluc <- runif(1, -0.02, 0.02)  
  default_rate <- 0.02 + default_rate_fluc  
  balances <- sim_average_return(  
    interest_rate = 0.03, default_rate = default_rate, loans = 3000)  
  balances  
}  
  
returns_with_fluc <- replicate(1e5, sample_return_when_default_rate_fluc())  
returns_flat <- data.frame(return = returns_with_fluc)  
  
returns_flat |>  
  ggplot(aes(return)) +  
    geom_density(fill="blue", alpha=0.2) +  
    geom_vline(xintercept=0., lty="dashed") +  
    xlab("return (interest rate = 3%, 3000 loans)") +  
    theme_minimal()
```

Loss rate < 1% if default rate is fixed



Loss rate ~ 34%

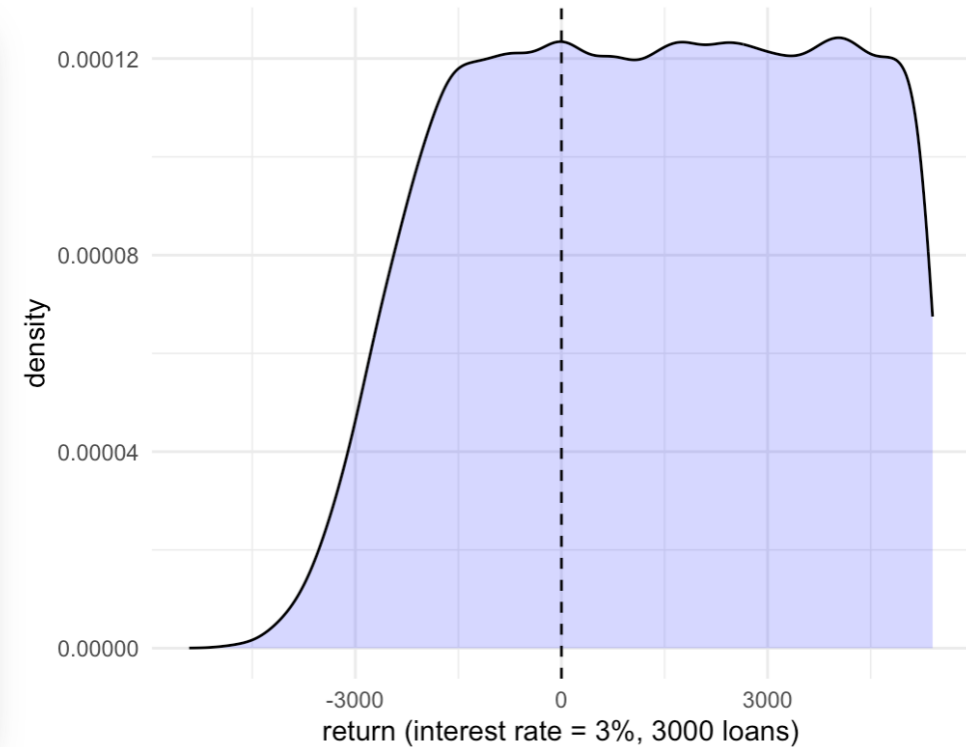
```

sample_return_when_default_rate_fluc <- function(){
  default_rate_fluc <- runif(1, -0.02, 0.02)
  default_rate <- 0.02 + default_rate_fluc
  balances <- sim_average_return(
    interest_rate = 0.03, default_rate = default_rate, loans = 3000)
  balances
}

returns_with_fluc <- replicate(1e5, sample_return_when_default_rate_fluc())
returns_flat <- data.frame(return = returns_with_fluc)

returns_flat |>
  ggplot(aes(return)) +
    geom_density(fill="blue", alpha=0.2) +
    geom_vline(xintercept=0., lty="dashed") +
    xlab("return (interest rate = 3%, 3000 loans)") +
    theme_minimal()

```



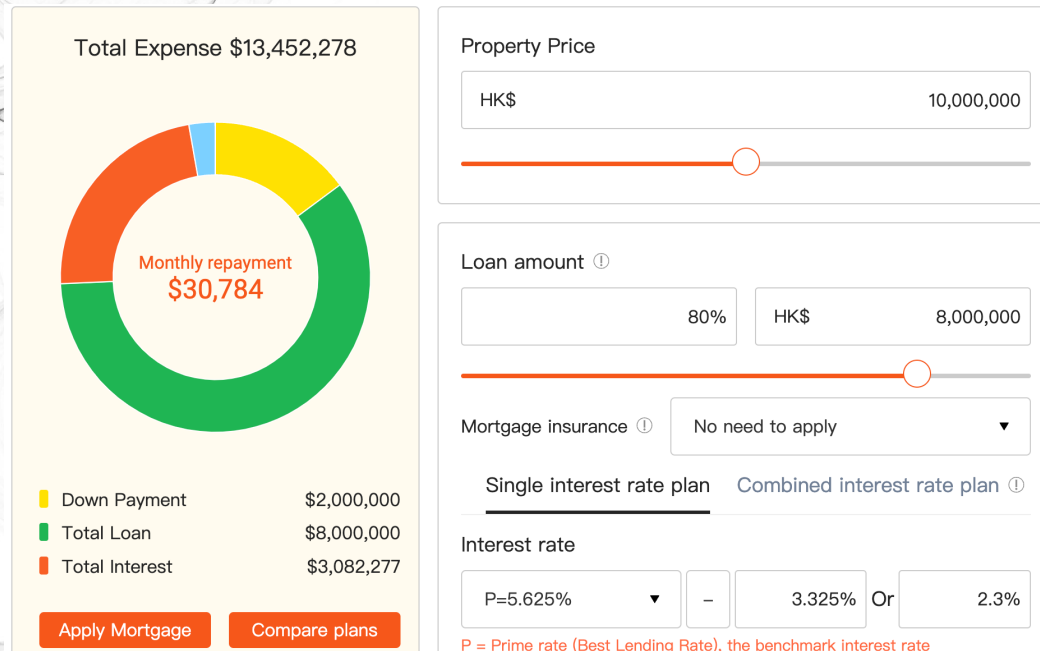
- This is not normal?
- Does it satisfy CLT?

More factors

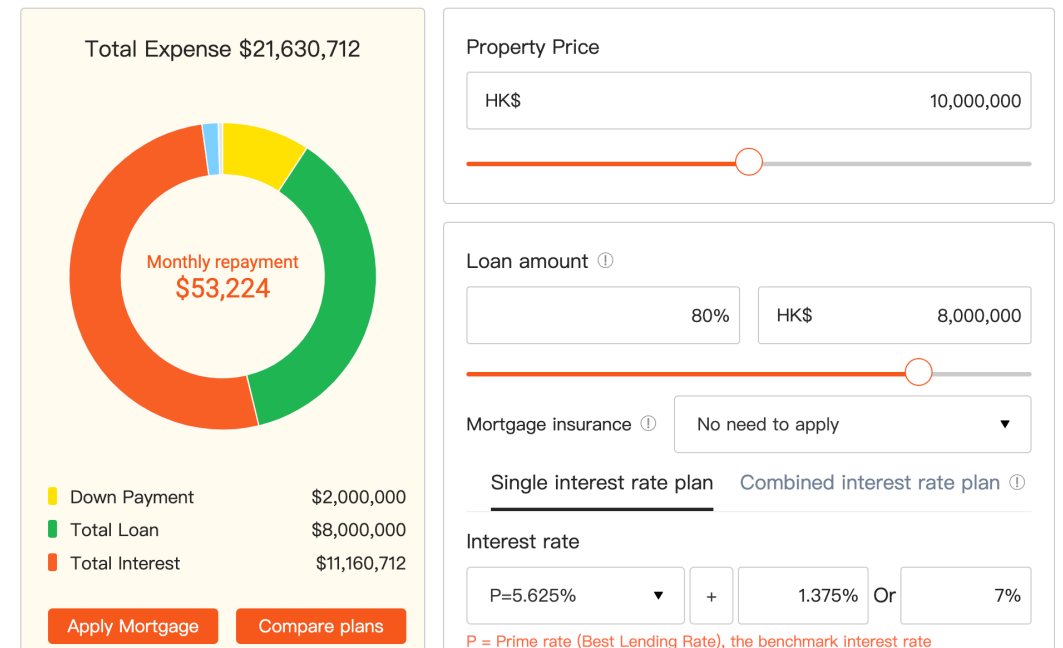
- Higher interest rate leads to higher default rate

For example: a 10M HKD property paid over 30 years

@2.3% interest rate
30,784 HKD monthly repayment



@7% interest rate
53,224 HKD monthly repayment



More factors

- Macro-economic changes & Black swan events
- Adjustable-rate mortgages
- Complex financial products that hide risks
- Cascading defaults

Clips from the 2015 movie “The Big Short”

- <https://www.youtube.com/watch?v=epb98OcFLZE>
- <https://www.youtube.com/watch?v=xbiDrzTd8fE>
- <https://www.youtube.com/watch?v=A25EUhZGBws>

Some explanations:

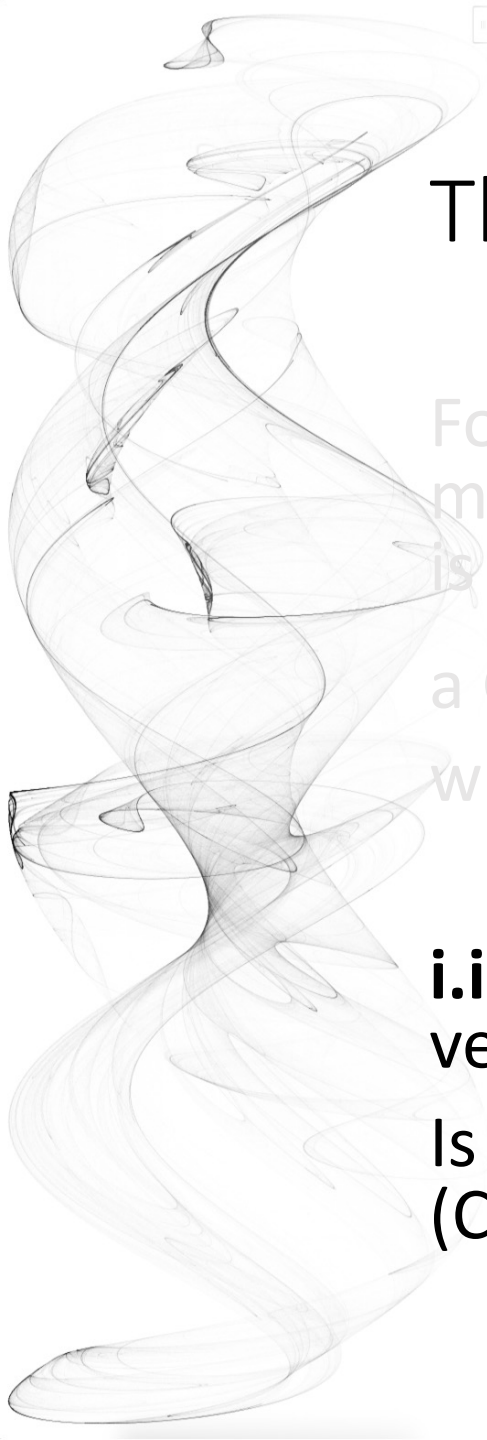
- <https://www.youtube.com/watch?v=zZZb8GhsEfc>





In this lecture

- Probability and random numbers
- Discrete probability
- Independence and conditional probability
- Continuous probability: expected value and variance
- The Law of Large Numbers, Central Limit Theory
- Case study: bank loan
- **The debate between Markov and Nekrasov**



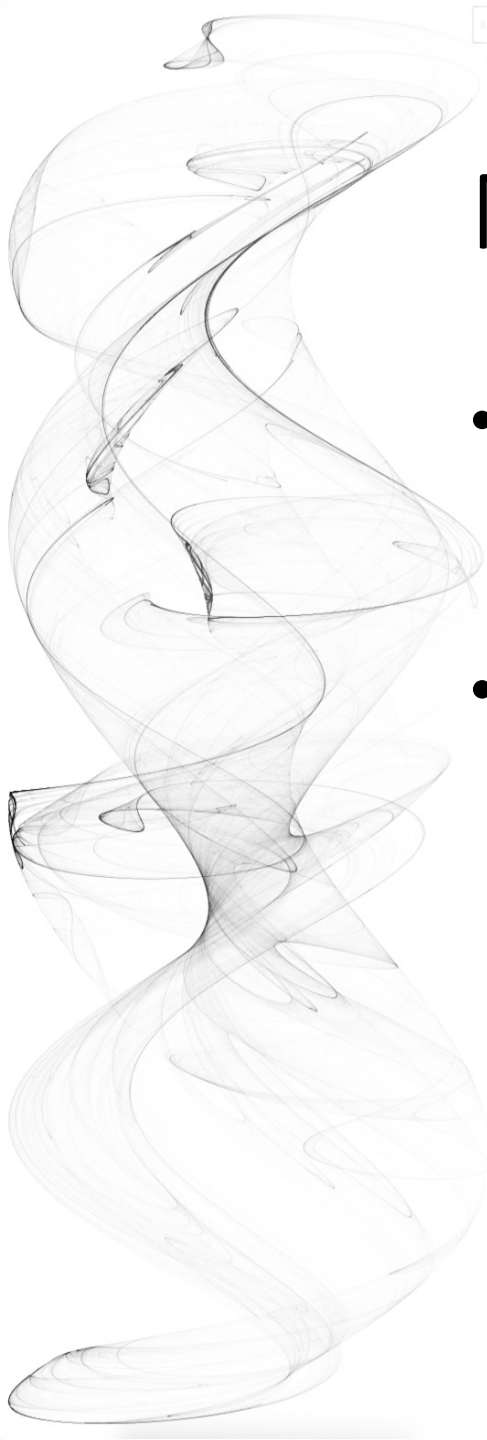
The debate between Markov and Nekrasov

For random variable X that follows **ANY** distribution (with finite mean and variance), **sample mean** of n **i.i.d.** observations (this is also a random variable) will converge to:

a Gaussian distribution with $E(X)$ as mean, and $\frac{SD(X)}{\sqrt{n}}$ as sd, when **n is large**.

i.i.d.: independent and identically distributed, this is usually a very strong assumption

Is it necessary? Can we get convergence (LLN) and normality (CLT) without i.i.d.?



Nekrasov's view

- Independence is a NECESSARY condition for LLN/CLT
- If you observed normal distribution, then the underlying data generation process is independent
 - If exam score distribution converges to a normal, then every participating student's mid-term performance is independent
 - Have you asked your classmates about midterm preparation?



Markov's counter example

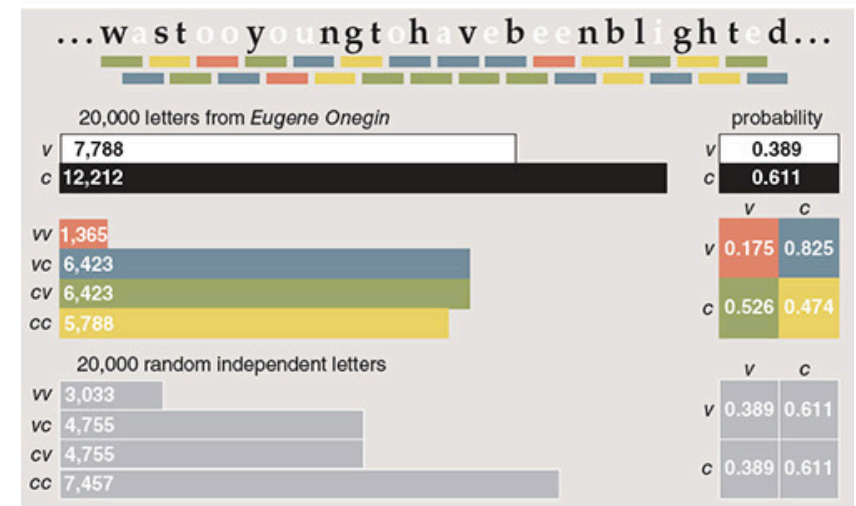


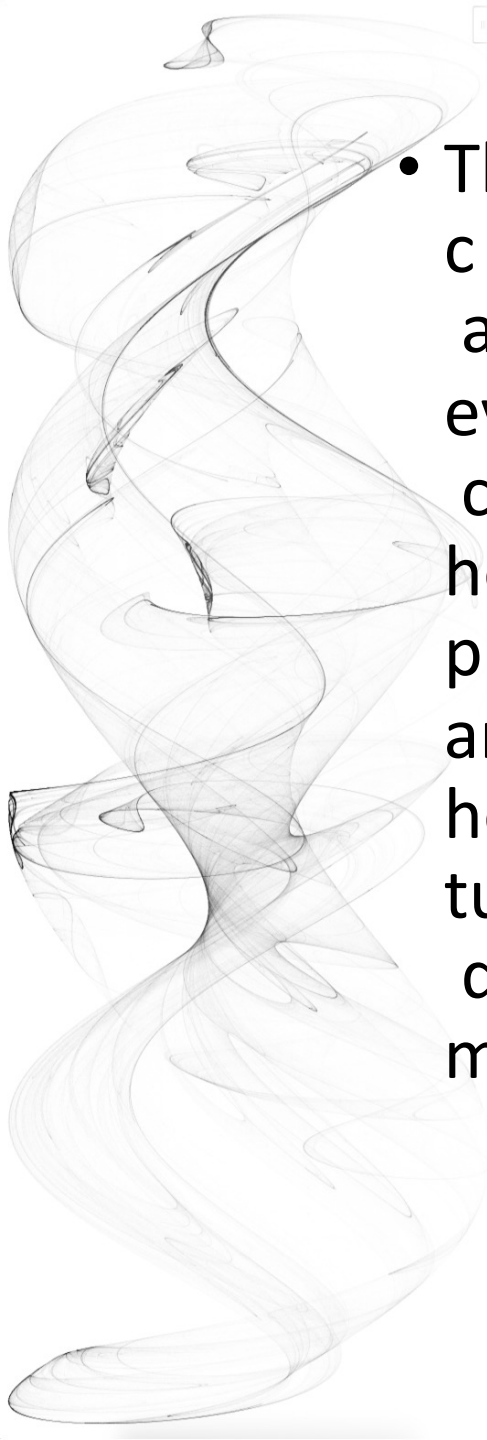
A. A. Markov (1886).

- In a free-form text, every letter is dependent on its previous letter(s).
- Analysis on “Eugene Onegin” by Alexander Pushkin

*He was too young to have been blighted
by the cold world's corrupt finesse;
his soul still blossomed out, and lighted
at a friend's word, a girl's caress.
In heart's affairs, a sweet beginner,
he fed on hope's deceptive dinner;
the world's éclat, its thunder-roll,
still captivated his young soul.
He sweetened up with fancy's icing
the uncertainties within his heart;
for him, the objective on life's chart
was still mysterious and enticing—
something to rack his brains about,
suspecting wonders would come out.*

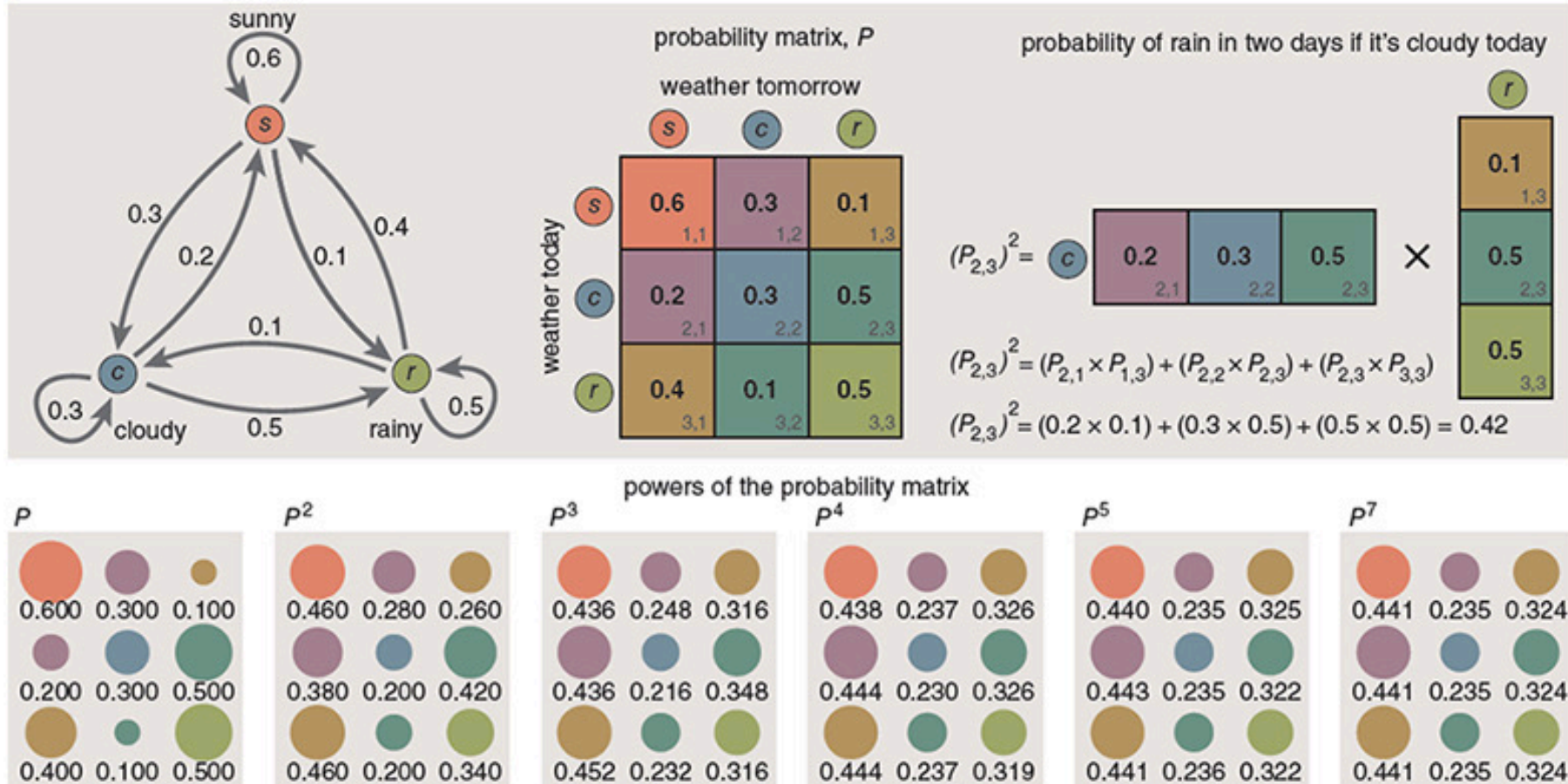
- Even the observations are all dependent, they still follow LLN.
- *Free will (independence) is not necessary to do probability*

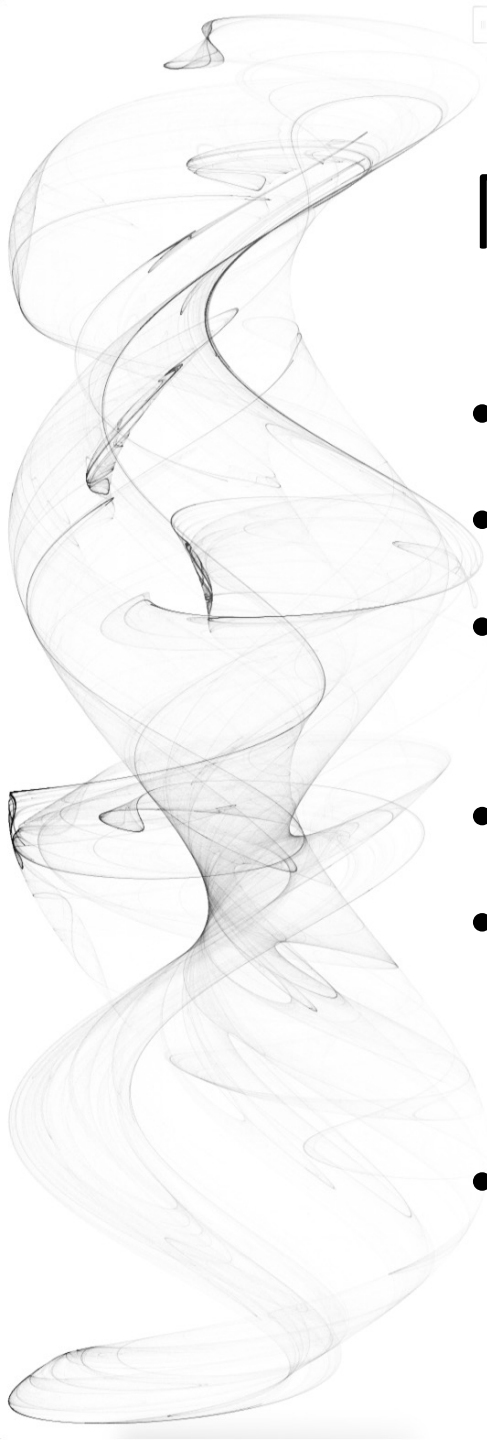


- 
- The debate between Markov and Nekrasov was not just academic but also ideological and political. Markov, a socialist and atheist, argued that the law of large numbers could apply to dependent events, proving that free will was not necessary for probability. In contrast, Nekrasov, a Tsarist and deeply religious, believed that the law of large numbers implied independence, which in turn implied free will. Their disagreement over the law of large numbers and the implications of independence led to a significant shift in the understanding of probability and the role of free will in the natural world. Markov's work on Markov chains and the subsequent development of the Monte Carlo method laid the foundation for modern statistical analysis and machine learning algorithms.

<https://www.blopig.com/blog/2021/09/chained-or-unchained-markov-nekrasov-and-free-will/>

Markov chain





Markov chain

- Monte Carlo simulation
- PageRank (Google)
- Hidden Markov Models for DNA/protein sequence analysis
- Markov Decision Processes
- Language models
 - Model how the current letter is dependent on its previous letters (context window)
- Many, many more



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